

Virtualization Technology



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Storage Virtualization in Cloud Computing

Introduction

- Rapid computer automation and Virtualized IT environments has been demanding additional resources in terms of storage, networking, and computing.
- Datacenters are trying to adopt virtualization in all possible ways to achieve this ever--increasing demand.
- Virtualization enable businesses to run multiple OS concurrently and abstracting physical data storage resources to make them appear as if they were a centralized resource (**Storage Virtualization**).
- Virtualization and its benefits are adopted in multiple domains from different server platforms and different hypervisors, to **storage** and even the network.

Introduction : Sources of Cloud Data

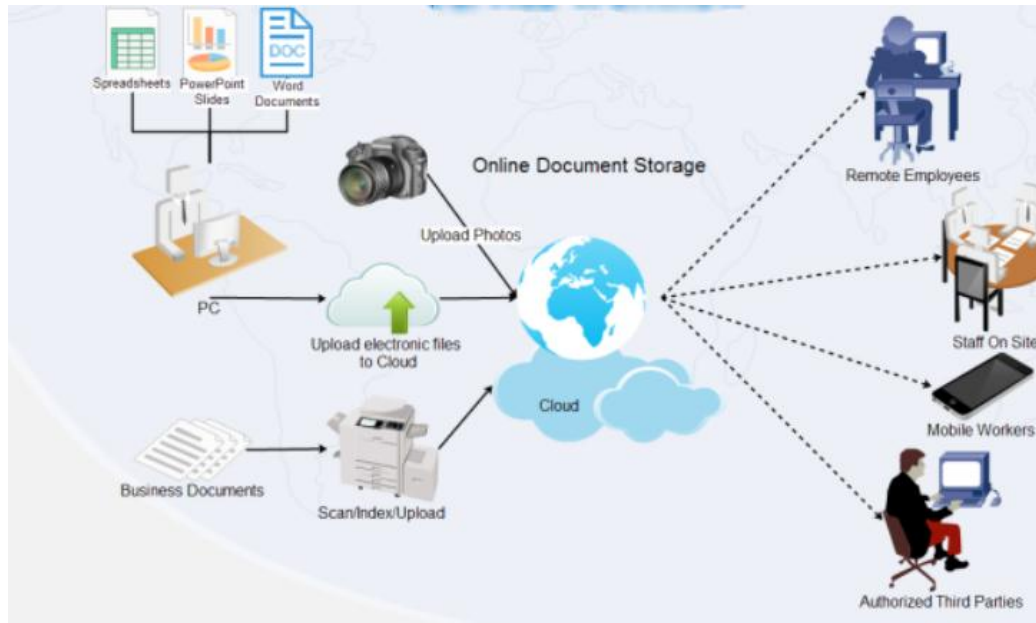
- The huge data that is generated by organizations in today's business environment has been focused on the 3V's –
 - Volume of data
 - Velocity of data
 - Variety of data
 - **Variability of data** (Increase in the range of values)
 - **Value of data** (Need for valuation)

Sources of data

- The bulk of data generated comes from three primary sources :
 - Social Data : Social Media Platforms (Likes, Tweets, Comments, Video Uploads)
 - Machine Data : Sensor Data (Industrial Equipment Sensor, Weather Sensor, IOT Devices)
 - Transactional Data : Routine online transaction data (Invoices, payment orders, storage records, delivery receipts)

Introduction : Cloud Storage

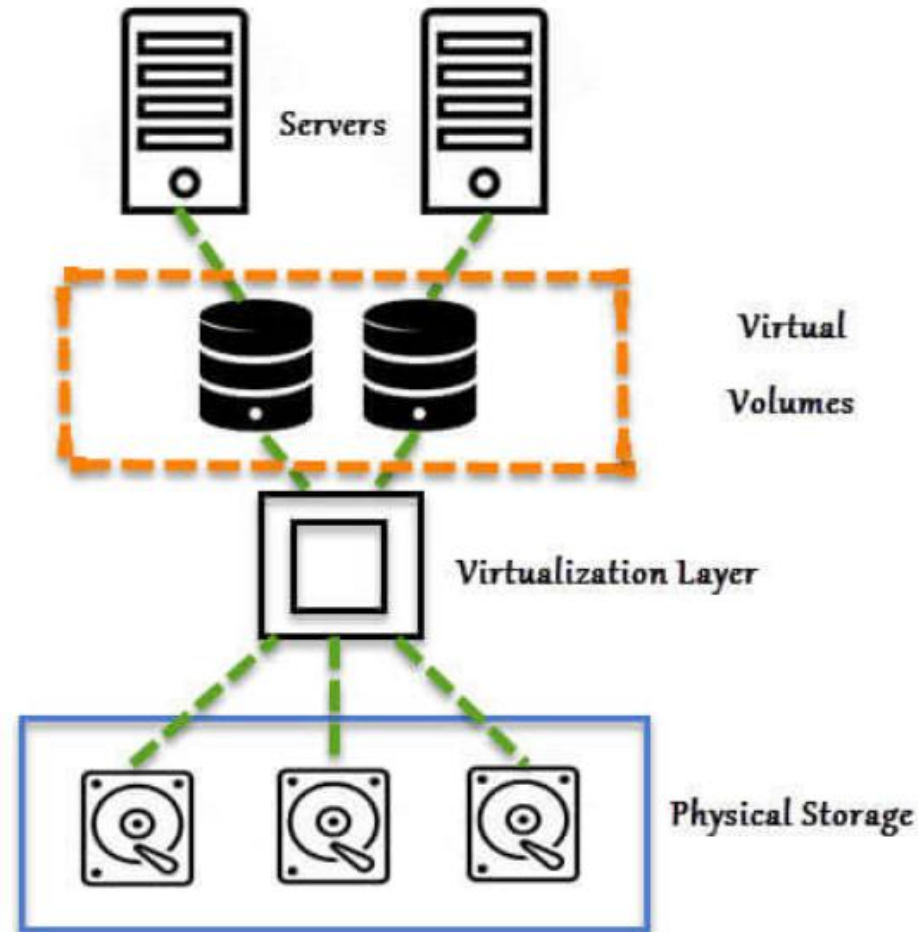
- Storage (Data storage) is the recording (storing) of digital information (data) within a digital data storage device using computer internet technology
- Data storage is a mechanism that enables a computing device to retain data, either temporarily or permanently.



Storage Virtualization : An Introduction

- Process of presenting a **logical view** of physical storage resources to hosts
- Storage virtualization is a system administration practice that allows **decoupling the physical organization of the hardware from its logical representation**.
- Sharing of physical storage into multiple storage devices which further appears to be a single storage device.
- Allows to harness a wide range of storage facilities and represent them under a single logical file (managed from a central console).
- Addresses the storage and data management problem by facilitating easy backup, archiving and recovery tasks by consuming less time.
- Storage virtualization aggregates the functions and hides the actual complexity of the storage area network (SAN).

Storage Virtualization : An Introduction



Why Storage Virtualization should be Implemented?

- As per Aberdeen survey, when businesses virtualize their storage, following benefits can be achieved:
 - Reduce hardware and resource costs :
 - 35% reduced overall IT expenses
 - Improved storage management in a heterogeneous IT environment
 - Improve reliability and performance
 - Reduction in application downtime
 - Constant access to applications, providing load balancing and failover across systems and sites.
 - Rapid recovery in case of an unexpected failure
 - Improve agility and ability to scale
 - Provides scaling storage to small and midsize companies
 - Resources are available at different times for one application or another.

Benefits of Storage Virtualization

- Data is stored in a very convenient location (Incase of host failure data don't get compromised necessarily).
- By using storage level abstraction, data storage becomes flexible i.e. how storage is provided, protected, partitioned and used.
- Storage Devices can facilitate advanced functions such as disaster recovery, duplication, replication of data & re-duplication of data.

Configuration of Storage Virtualization

- **In-band method :**

- The virtualized environment configuration is stored in the data path
- Occurs when the virtualization engine operates between the host and storage.
- Both I/O requests and data pass through the virtualization layer, which allows the engine to provide advanced functionality like data caching, replication and data migration.
- Takes up fewer host server resources, as it does not have to find and attach multiple storage devices

- **Out of band method :**

- Virtualization implementation takes place external to the data path.
- Splits the path into control (metadata) and data paths
- Installs agents on individual servers to direct their storage I/O to the virtualization appliance

Storage Virtualization Risks



How Storage Virtualization Apply?

- **Host-Based Storage Virtualization :**

- Refers to the way a virtualization host presents storage to a guest OS.
- Virtualizations and management is done at the host level with the help of software and physical storage, it can be any device or array.
- The host is made up of multiple hosts which present virtual drives of a set to the guest machines.

- **Network-Based Storage Virtualization :**

- Devices such as a smart switch or purpose-built server connect to all the storage devices through a optical fibre channel storage area network (SAN) and present the storage as a virtual pool.

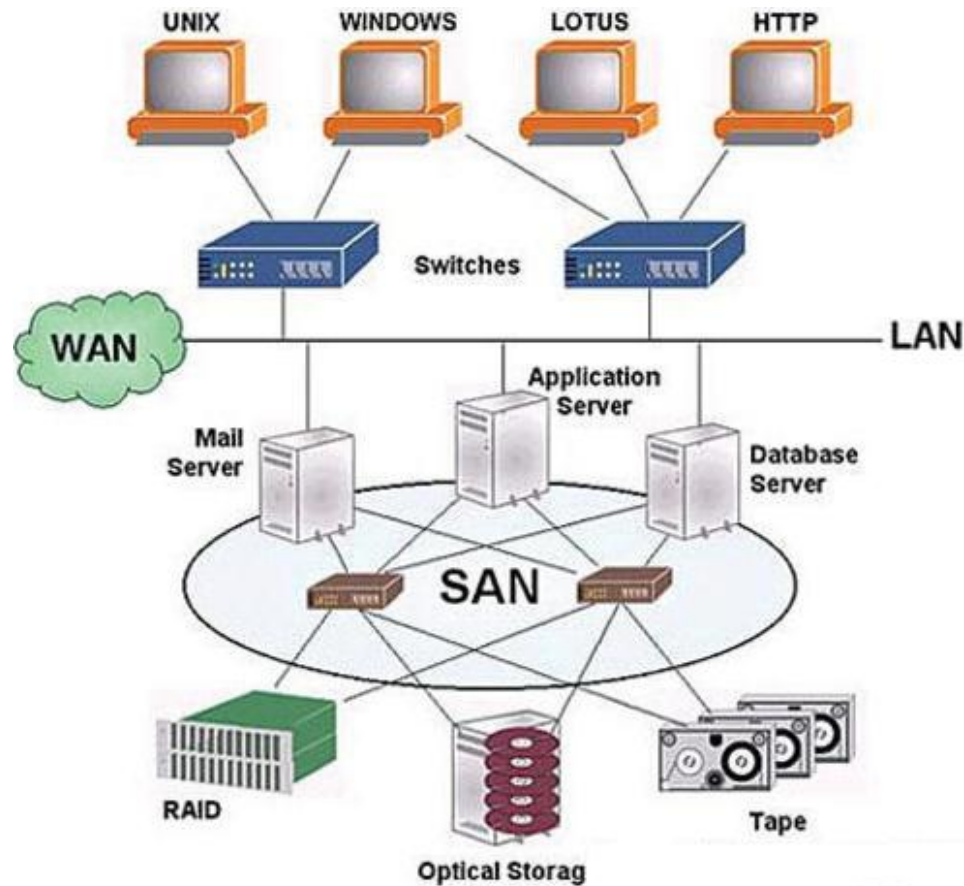
- **Array-Based Storage Virtualization :**

- Refers to hardware-level storage tiering.
- Available software handles the amount of storage tier made up of Solid-State Drives (SSD).

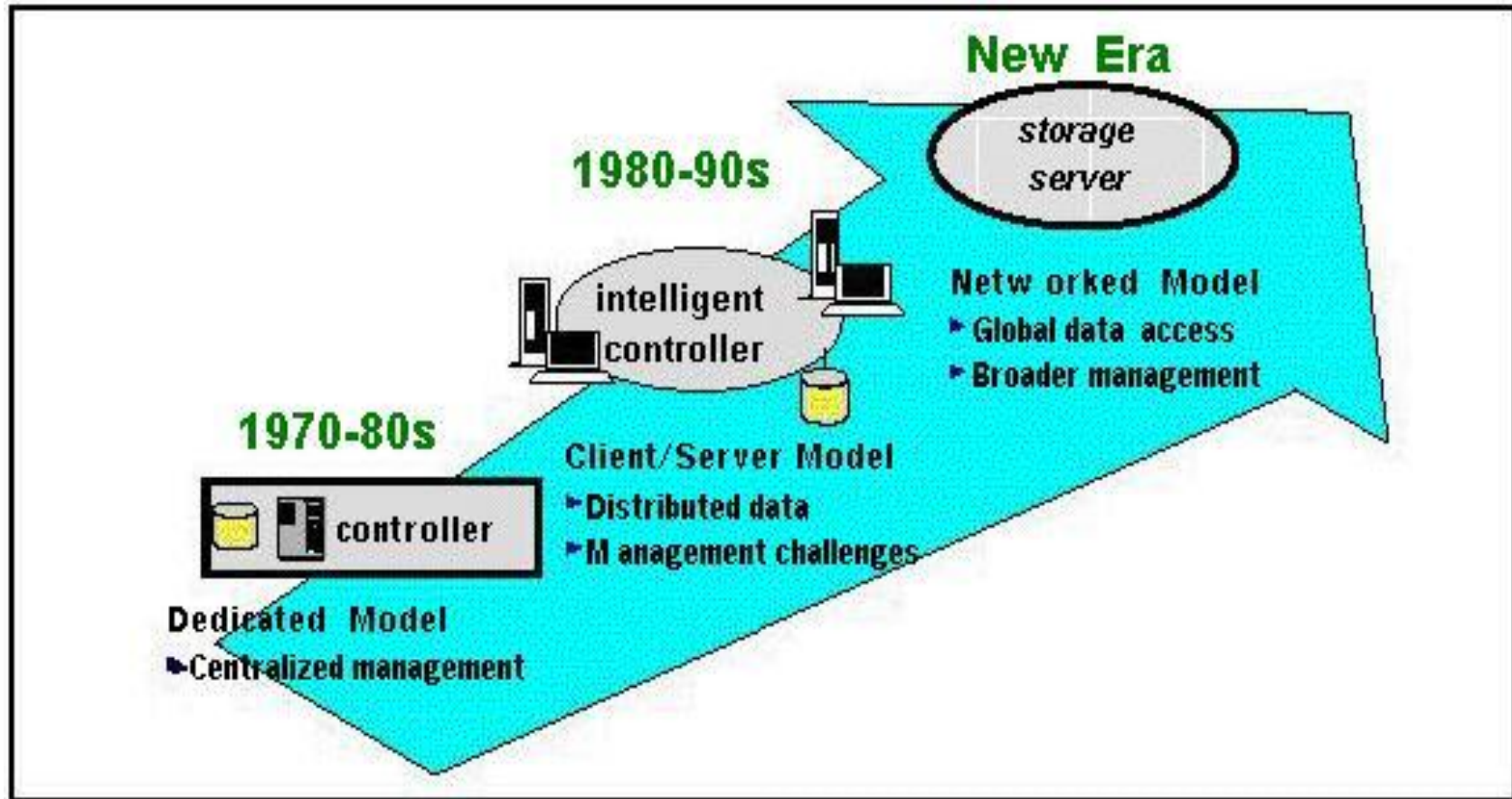
Storage Area Network

- A network of storage devices that can be accessed by multiple servers or computers, providing a shared pool of storage space.
- A dedicated high-speed network that interconnects and presents shared pools of storage devices to multiple servers.
- Typically composed of hosts, switches, storage elements, and storage devices that are interconnected using a variety of technologies, topologies, and protocols.

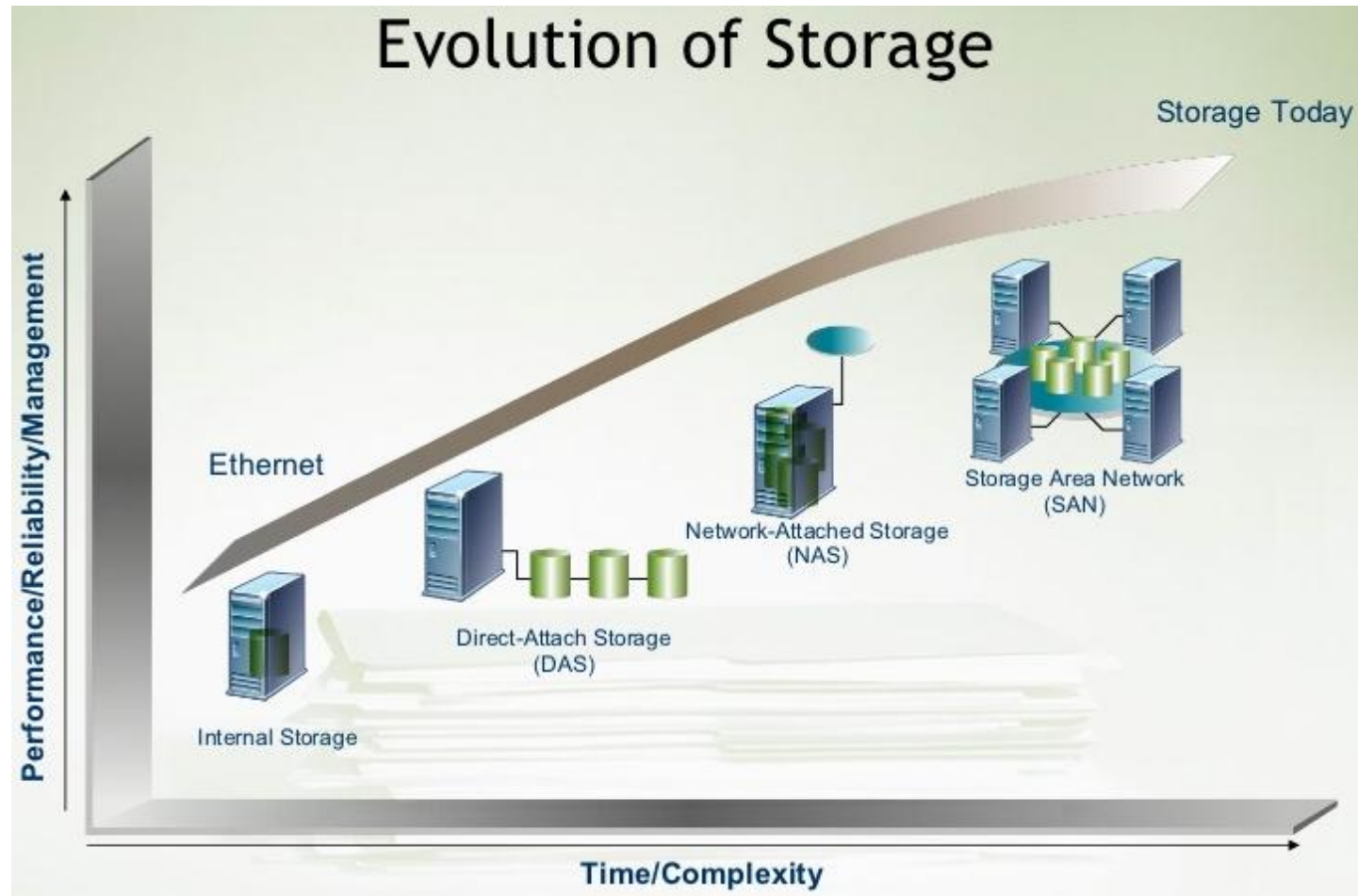
Storage Area Network



Evolution in Storage Architecture



Evolution of Storage Area Network



Problems with Traditional Storage Methods

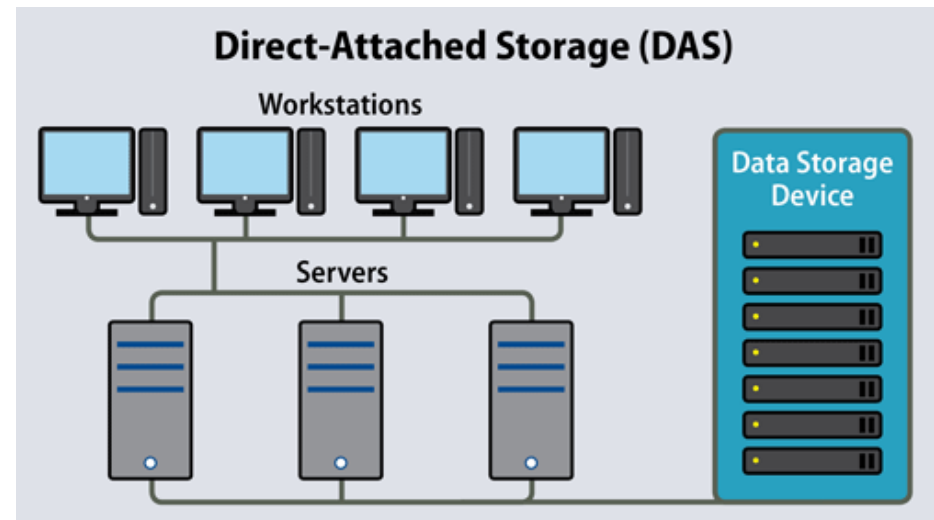
- Scalability --Rapidly growing data volume
- Connectivity --Distributed data sharing
- Requirement of 24/7 storage availability,
- Non permissible single point failure
- High performance
- Easy management

Direct (Server) Attached Storage

- Digital storage that is attached directly to a computer or a server.
- Typical DAS system is made of a data storage devices connected directly to a computer through a host bus adapter (HBA).
- 1:1 relationship exists between storage and host
- There is no network devices (like hub, switch, or router) between the storage device and the computer.

Protocols for DAS connections

- Advanced Technology Attachment (ATA)
- Serial Advanced Technology Attachment (SATA)
- Small Computer System Interface (SCSI)
- Serial Attached SCSI (SAS)



Direct Attached Storage

- **Advantages**

- High availability.
- High access rate
- Elimination of network setup complications.
- Storage capacity expansion.
- Data security and fault tolerance.

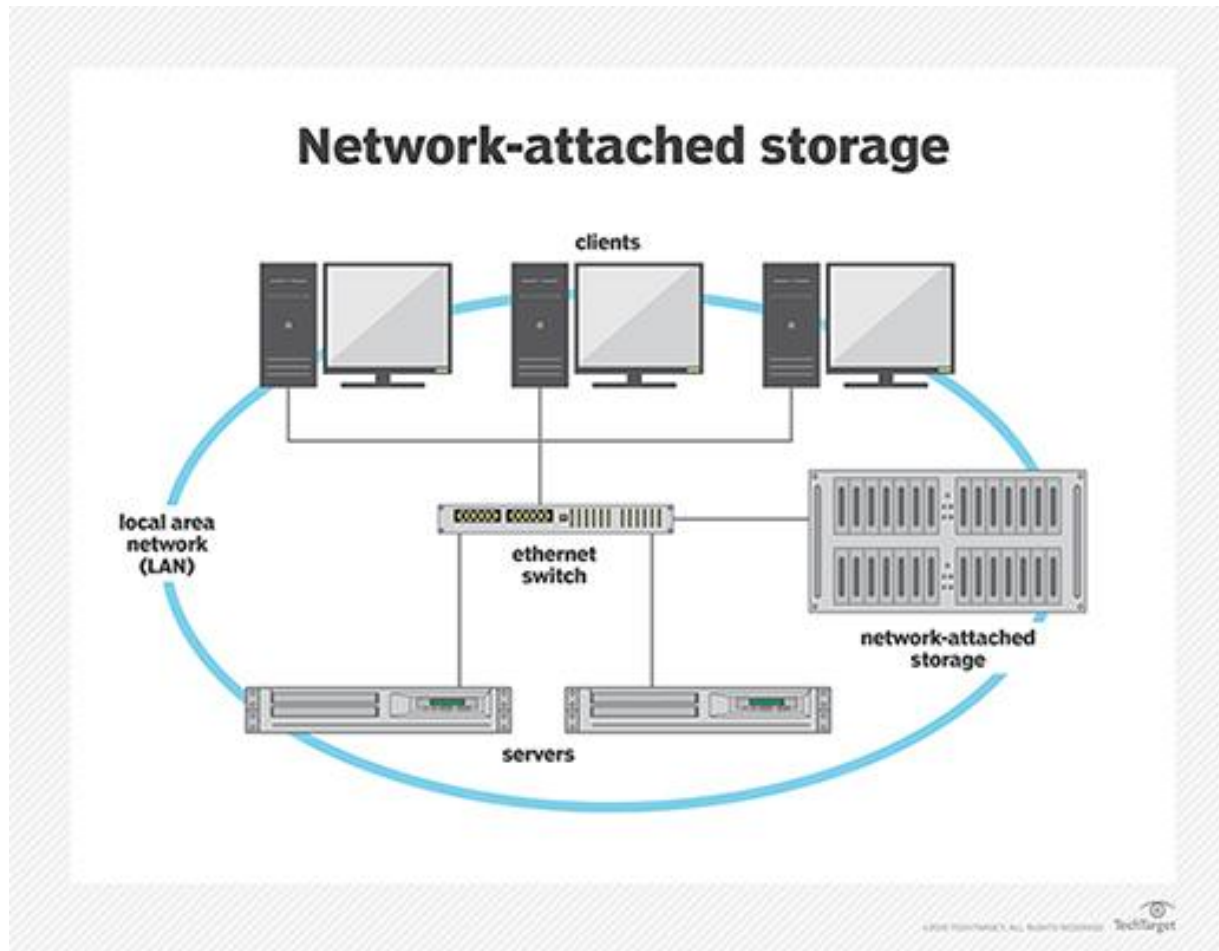
- **Disadvantages**

- Data not accessible by diverse user groups.
- Allows only one user at a time.
- High administrative costs.

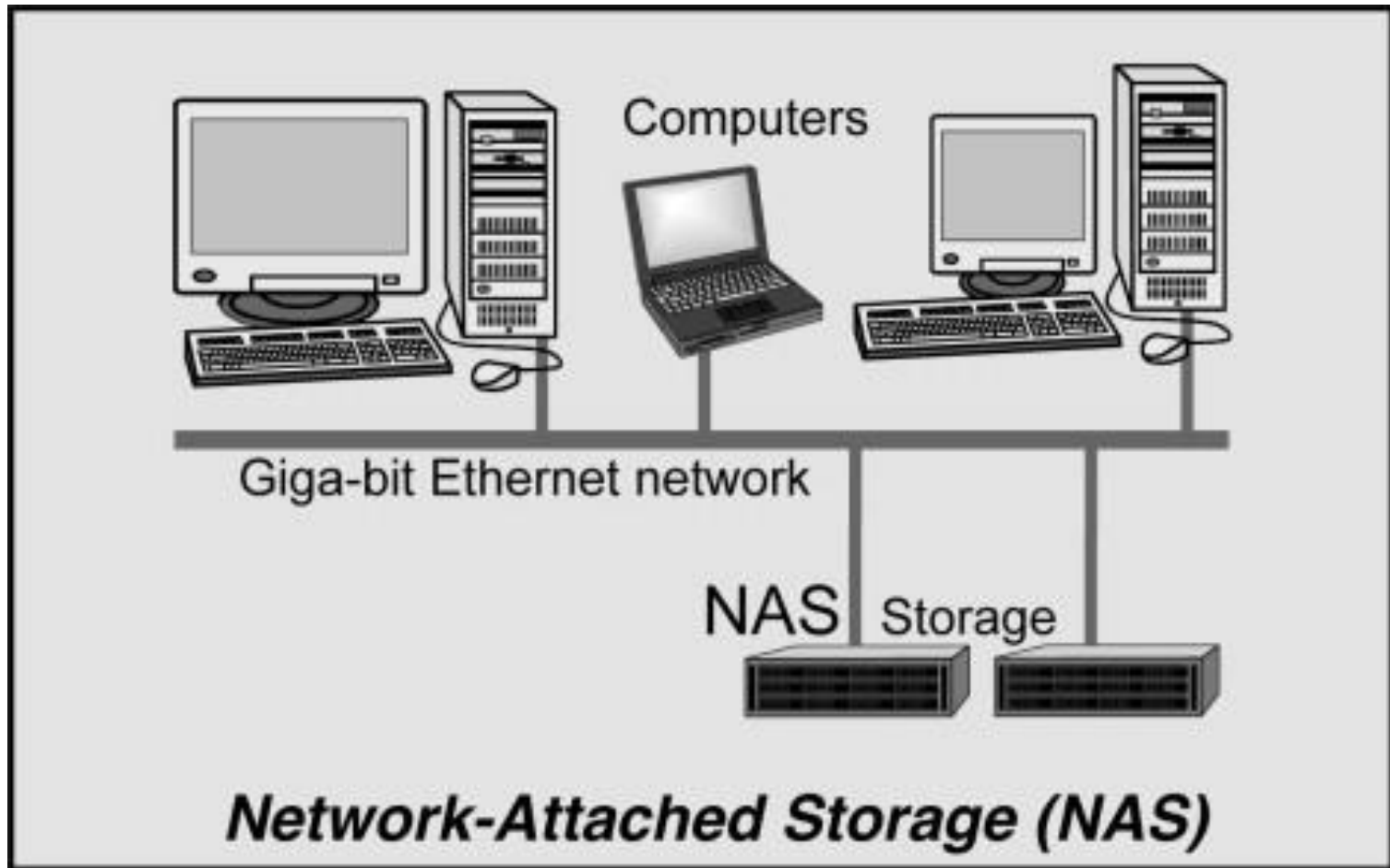
Network Attached Storage

- **Network-attached storage (NAS)** is a file-level computer data storage server connected to a computer network providing data access to a heterogeneous group of clients.
- Provides file-based data storage services to other devices (group of users) on the network
- A shared storage on a network infrastructure
- The files to be served are typically contained on one or more storage drives, often arranged into logical, redundant storage containers or RAID.
- The device itself is a network node, much like computers and other TCP/IP devices, all of which maintain their own IP address and can effectively communicate with other networked devices.

Network Attached Storage



Network Attached Storage



Benefits of Network Attached Storage

- Next step up for a home office or small business that is using DAS (direct attached storage)
- Relatively inexpensive
- Good expandability
- NAS facilitates
 - Redundant storage architecture
 - Sharing files locally and remotely
 - Flexibility
 - Having access to files 24/7
 - Ability to replace and upgrade hard drives in the system
 - Automatic backup.

Limitations of NAS

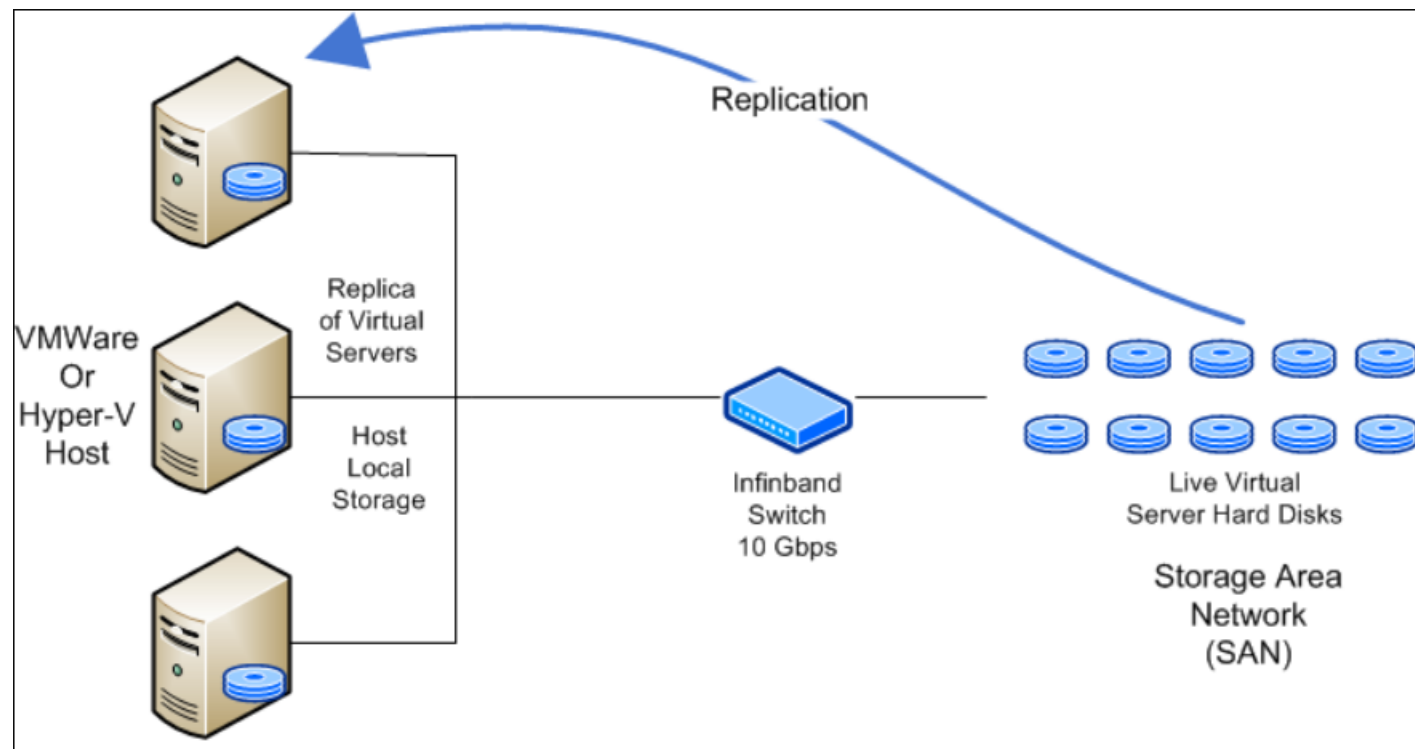


- Poor scaling performance (Require addition of servers for load balancing)
- Ethernet (Delayed packet delivery)
- Latency (slow or retried connections)

Storage Area Network

- A storage area network (SAN) is a dedicated high-speed network or subnetwork that interconnects and presents **shared pools of storage devices** to multiple servers.
- Provide users shared access to consolidated, block level data storage, even allowing multiple clients to access files at the same time with very high performance.
- Provides the highest access speed available for media and mission critical stored data.
- **A SAN is typically employed to consolidate storage**

Storage Area Network



Importance of Storage Area Network

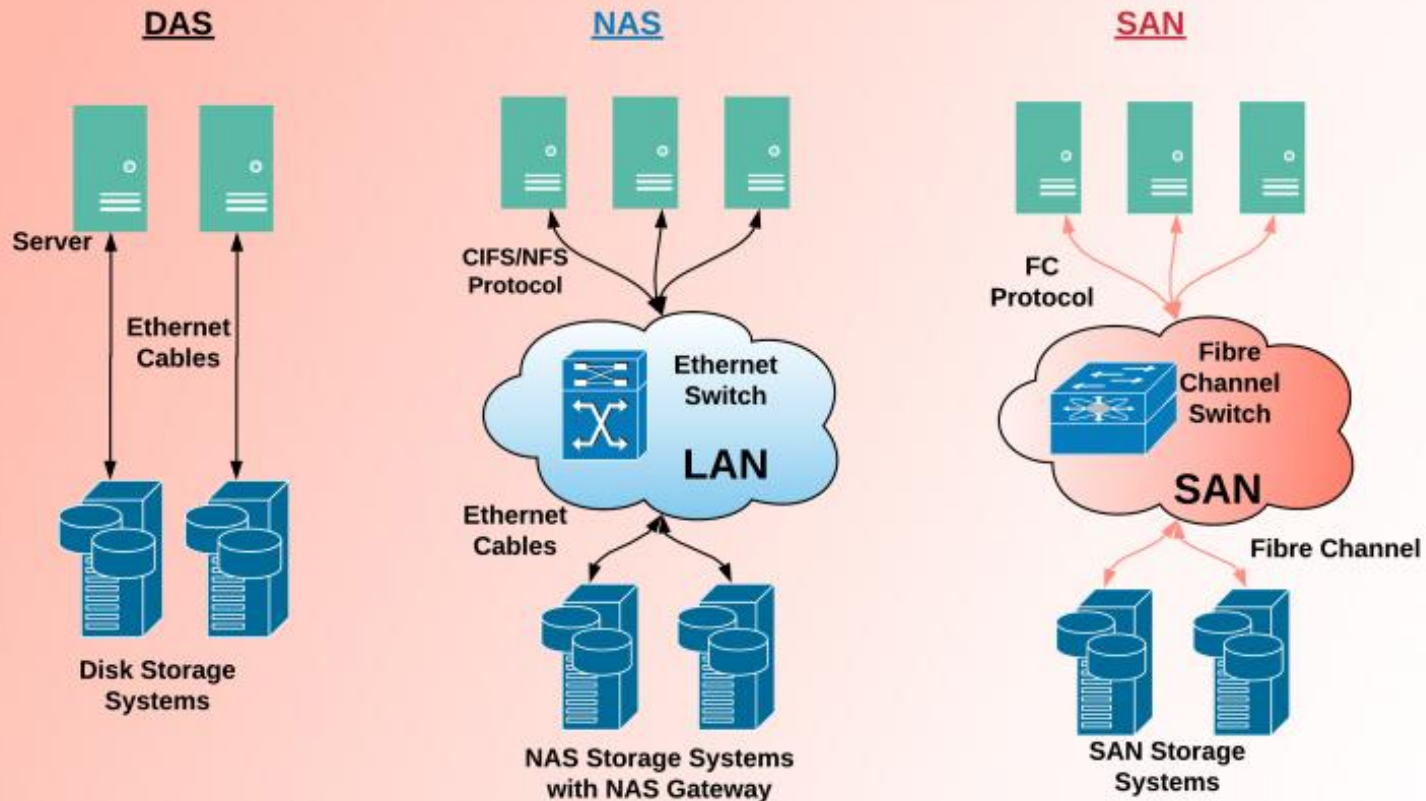
- A server generally includes one or more local storage devices.
- Considering a data center with hundreds of servers, each running VMs that can be deployed and migrated between servers as desired.
- If data for one workload is stored on that local storage, the data might also need to be moved if the workload is migrated to another server or restored if the server fails.
- **Possible Solutions**
 - Take necessary steps to organize, track and use the physical disks located in individual servers throughout the data center
 - Move storage to a dedicated storage subsystem, i.e. storage array, where the storage can be collectively provisioned, managed and protected (i.e. **Storage Area Network**).
- SAN is essentially a network fabric of interconnected computers and storage devices.

Features of Storage Area Network

- Improves storage availability
- Uninterrupted storage service (Availability of alternative path through the SAN fabric)
- Improves storage utilization (Eliminates "forgotten" disks on underutilized servers)
- Offers a central location for all storage
- Provides ability
 - To support enterprise workloads
 - To handle disaster recovery
 - To improve business continuity postures
 - Enhance the organization's regulatory compliance

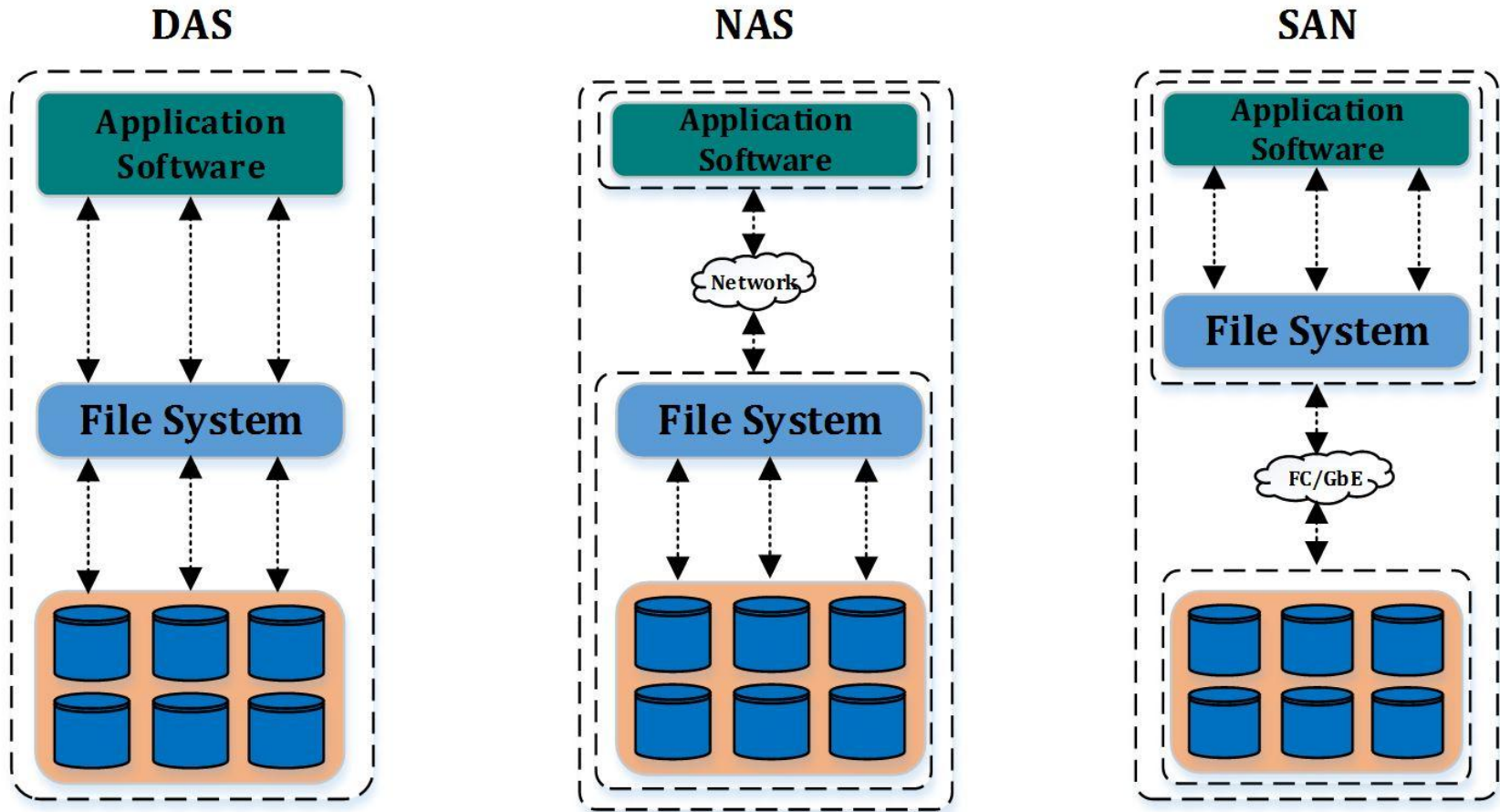
Storage Architecture Comparison

DAS vs NAS vs SAN Architecture



www.mvcloudwiki.com

Storage Architecture Comparison



Storage Architecture Comparison

Storage Structure	Network	Connection Media	Input/Output Protocols	Bandwidth	Availability
DAS	No	Parallel SCSI / FC	SCSI	40 MBps up to 160 MBps	Low
NAS	Yes	Ethernet	NFS/CIFS	10 MBps to 1 Gbps	Moderate
SAN	Yes	Fiber Channel/Ethernet	SCSI	1Gbps up to 50Gbps	High

NAS vs. SAN

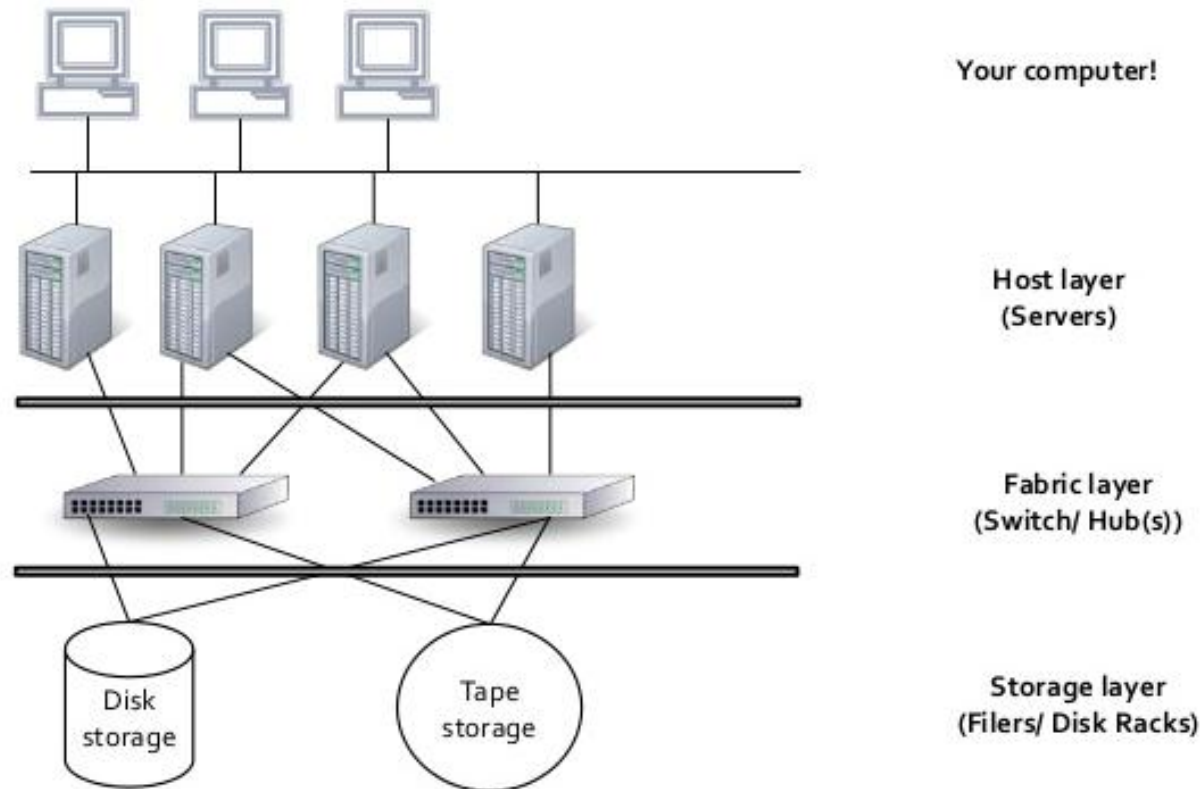
	NAS	SAN
Fabric	Uses TCP/IP networks, most commonly Ethernet	Runs on high speed Fibre Channel networks
Data processing	Processes file-based data	Processes block data
Protocols	Connects directly to an Ethernet network Can use several protocols to connect with servers including NFS, SMB/CIFS, and HTTP	Uses SCSI protocol to communicate with servers
Performance	Generally has lower throughput and higher latency because of its slower file system layer	A higher performer for environments that need high-speed traffic
Scalability	Entry level and NAS devices are not highly scalable, High-end NAS systems scale to petabytes using clusters or scale-out nodes	Scalability is a major driver: its network architecture enables admins to scale performance and capacity in scale-up or scale-out configurations
Ease of management	Easier to manage: device easily plugs into the LAN and offers a simplified management interface	Requires more administration time than NAS
Price	In general NAS is less expensive to purchase and maintain, although a high-end NAS will cost more than an entry-level SAN	SANs are more complex to manage with FC SANs on top of the complexity heap

Working of SAN

- SAN is essentially a network that is intended to connect servers with storage.
- The goal of SAN is to take storage out of individual servers and locate the storage collectively for centralized management and protection.
- Centralization can be performed physically, i.e. placing disks into a dedicated storage subsystem like a storage array.
- A better way to implement centralized storage is through a logical approach using virtualization software's.
- Connecting the collective storage to servers through a dedicated fiber network improves storage network traffic performance.

Designing the SAN

- A SAN is generally represented as a series of three distinct layers : a host layer, a fabric layer and a storage layer.



Designing the SAN

Host Layer :

- The servers reside in this layer.
- Servers run the high-performance applications that run our business applications.
- Every server in the host layer uses a Host Bus Adapter (HBA) to connect it to the fabric layer.
- The HBA connects the server with a fiber-optic cable to the hubs or switches in the fabric layer.

Fabric Layer :

- Actual network part of the SAN and is also known as SAN layer
- It's the central connection point between the servers in the host layer and the storage devices in the storage layer.
- Devices in the fabric layer enables SAN to allow flow of data between the host layer and the storage layer.

Designing the SAN

Storage Layer :

- Comprises of various types of storage devices consolidated into various storage pools, tiers or types.
- Storage devices within a SAN are organized into physical RAID groups that can be employed to increase storage capacity, improve storage device reliability or both.



THANK YOU